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Pre-Arrival Confidence and Perceived Importance in First-Year UK Sport Students: A Multi-Institutional Examination of Gender, Institution and Programme Differences

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Abstract

This multi-institutional study examined pre-arrival confidence and perceived importance among first-year sport students across three post-92 universities and one public research university exploring programme of study, gender, and institutional differences, while also evaluating the psychometric properties of the pre-arrival survey. Of 1033 eligible students, 604 (58%) completed the survey across 25 sport-related programmes grouped into six categories: physiotherapy and rehabilitation, sport and exercise science, sports exercise and nutrition, sports coaching, sports management, and sport foundation year. Psychometric validation of the pre-arrival survey demonstrates its reliability and validity, providing the sector with a robust, standardised tool for assessing incoming students' preparedness. Significant programme differences include physiotherapy and rehabilitation students reporting higher learning confidence, learning importance, and community confidence compared to other programmes. Female students demonstrated significantly higher learning importance and health and well-being importance than male students, though no gender differences in confidence were observed. Institutional variation was minimal, with one institution showing higher learning importance. Socioeconomic indicators did not significantly influence pre-arrival responses. The findings highlight the need for differentiated pre-arrival support targeting programme-specific confidence gaps and gender-related differences in perceived importance. The validated PAS provides a reliable tool for early identification of students requiring enhanced transitional support, potentially addressing persistent retention and progression challenges in UK sport programmes.



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1. Introduction

A successful transition to higher education¹ (HE) involves students building confidence, forming relationships with staff and peers, staying motivated, and feeling a sense of belonging ([Meehan and Howells 2019](#); [Pedler et al. 2022](#); [Thompson et al. 2021](#); [Nallaya et al. 2022](#)). Studies show that students face complex challenges moving from sixth form or college (SFC) to HE, and unmet academic or social expectations can lead to

disengagement, poor performance, or withdrawal (Fellingham et al. 2024; Gravett and Winstone 2021; Taylor and Harris-Evans 2018; Smith et al. 2022; Williams and Roberts 2023; Strayhorn 2023; Gillen-O'Neel 2019; Turner et al. 2017; Young et al. 2020). New students entering higher education from SFC settings often lack confidence in meeting academic standards (Gill 2019; Morgan 2023). They report concerns about employability, self-regulation, workload, independent learning, degree expectations, collaborative and critical learning, and balancing study with work (Hayman et al. 2020; Jonker et al. 2011; Pather and Dorasamy 2018; Rowley et al. 2008; Hockings et al. 2018; Farhat et al. 2017; Lowe and Cook 2003; Hassel and Ridout 2018).

Over the past two decades, sport degree programmes in the United Kingdom have expanded significantly and are now available at 125 universities with a total of 639 courses (Universities UK 2024). Despite this growth, challenges persist related to student retention and progression (Timmis et al. 2022; Tomlinson et al. 2024). For instance, only 74% of students graduated from these programmes in 2022, which falls short of the sector average of 85% (HESA 2024). Research by Hibbs et al. (2024) and Timmis et al. (2024) highlights a gap between the teaching approaches used in SFC and HE environments, often noting that university instruction is less directive and more reliant on student autonomy. Additionally, Hayman and colleagues (Hayman et al. 2017, 2020) identified that first-year sport students frequently begin higher education feeling uncertain about meeting academic standards, establishing social connections, adapting to living independently, and developing self-guided learning skills. This uncertainty may be heightened for first year students starting university from a non-traditional background (i.e., not a direct school leaver) and with a growing number of students using this route (Hubble and Bolton 2021), the need for additional understanding and support has never been greater.

Recent research has investigated demographic variations in pre-arrival confidence and perceived importance regarding learning, community, employability, and health and well-being factors among sport students, utilizing the Pre Arrival Survey (PAS; Evasys 2024). Findings indicate that female sport students demonstrated higher levels of pre-arrival confidence in forming friendships and joining societies, and placed greater emphasis on learning, health, well-being, and community participation compared to their male counterparts (Hibbs et al. 2025a). Engagement in sports and extracurricular activities has been found to enhance sport students' sense of belonging and acceptance (Hayman et al. 2022), while the comparatively lower prioritization of health and well-being by male students (Hibbs et al. 2025a) is noteworthy given its link to academic engagement and retention (Thomas et al. 2021). Gender differences in the transition to HE can be understood through the lens of academic integration theory (Tinto 1993) and help-seeking behaviour frameworks (e.g., Theory of Planned Behaviour; Ajzen 1991). Research demonstrates that gendered socialisation patterns influence how students approach academic challenges, seek support, and engage with university life (Brown et al. 2021). Male students, particularly in sport contexts, may be influenced by traditional masculine norms emphasising self-reliance and emotional restraint, which can affect their willingness to acknowledge struggles or seek help during transition periods.

Further analysis by Hibbs et al. (2025b) assessed whether pre-arrival confidence and the perceived importance across domains predicted academic achievement and retention within a cohort of 368 first-year sport students at an English post-92 university. The results revealed that females outperformed males academically across all programs, with physiotherapy students attaining the highest grades. Confidence and perceived importance in health and well-being were significant positive predictors of academic achievement, whereas the perceived importance of community was associated with an increased risk of withdrawal.

Students in all programmes of study face challenges transitioning to higher education, with these issues being especially evident among first-year undergraduate sports students (Timmis et al. 2024; Hayman et al. 2025). Programme-level differences may be explained in part by the vocational identity theory (Holland 1997), which states that students entering clearly defined professional programmes (such as physiotherapy) experience different transition processes than those in broader vocational fields (such as sport). Physiotherapy programmes have explicit career pathways and professional body requirements compared to broader sport-related degrees where career trajectories are more ambiguous and students may require additional qualifications for employability (Hall et al. 2019; Shen et al. 2024). These structural differences in programme characteristics may manifest as differential pre-arrival confidence and expectations, which subsequently influence engagement, achievement, and retention. Given that previous single-institution research has identified programme-specific patterns (Hibbs et al. 2025a, 2025b), multi-institutional investigation is necessary to determine whether these differences represent generalisable programme characteristics or institution-specific effects.

This study builds on Hibbs et al. (2025a) with the primary aim of investigating pre-arrival confidence and the perceived importance of university among first-year sports students at three post-92 universities and one public research university, focusing on gender and programme differences. In addition, there has not been a thorough evaluation of the PAS beyond a single study group or location. Hibbs et al. (2025a) is one of the few to assess this, finding the survey to be acceptably reliable and factorially sound for analysis. To ensure the PAS is effective across different institutions, it is crucial to confirm its reliability and factor structure in varied settings. Thus, a secondary aim of this research was to test the pre-arrival survey's measurement invariance and psychometric properties across different sport degree programs and institutions. This also involved examining the influence of socio-economic factors such as eligibility for free school meals, being the first in the family to attend university, and previous college qualifications. Confirming measurement invariance ensures the PAS consistently measures what it intends, regardless of institutional differences, which strengthens its usefulness for predicting important student outcomes in sport education—like retention, progression, academic performance, and dropout risk.

2. Method

2.1. Organisational Context

This cross-sectional study was conducted across four UK-based HE providers: three post-92 universities and one public research university². All four institutions (hereafter referred to as 1, 2, 3, and 4) were originally established to deliver high-quality vocational education, a focus that continues to shape their current provision. Each is research-active, socially inclusive, and recognised for teaching excellence, with strategic commitments to widening access and reducing educational inequality. All four institutions have a long and successful reputation of delivering industry-relevant sports-based degree provision across a range of disciplines, including science, coaching, management, nutrition, and physiotherapy.

2.2. Participants

From a pool of 1033 students meeting eligibility criteria, 604 individuals (representing 58% of the target population) provided complete survey responses. The sample comprised 354 males (59%) and 250 females (41%). The twenty-five distinct programmes offered across the participating institutions were consolidated into six broader categories for analysis purposes: Physiotherapy and Rehabilitation (P); Sport and Exercise Science (SES); Sports,

Exercise, and Nutrition (SEN); Sports Coaching (SC); Sports Management (SM); and Sports Foundation Year (FY). Table 1 displays the specific programmes available at each institution, while Table 2 presents the distribution of survey respondents by institution, gender, and programme category.

Table 1. Academic programmes offered across participating institutions.

Institution of Study			
1	2	3	4
Sport and Exercise Science	Sport and Exercise Science Health and Exercise Science	Sport and Exercise Science Applied Sport Science	Sport and Exercise Science
Sports Coaching	Community Football Coaching Sport and Youth Leadership	Sport Coaching and Physical Education	Sports Coaching and Performance Science
Sports Management			
Sport, Exercise, and Nutrition	Sport, Exercise, and Nutrition		
Physiotherapy		Sport and Exercise Therapy Strength and Conditioning with Rehabilitation	Sport Rehabilitation Physiotherapy
Sport Foundation	Sport Coaching and Development Sport and Exercise Science Community Football Coaching		Sports Coaching and Performance Science Sport and Exercise Science

Table 2. Distribution of participants by institutional affiliation, gender, and programme category (P = physiotherapy and rehabilitation; SES = sport and exercise sciences; SEN = sport, exercise, and nutrition; SC = sports coaching; SM = sports management; FY = sport foundation year).

Institution	Gender	P	SES	SEN	SC	SM	FY	Total
1	Male	16	68	5	34	30	32	297
	Female	34	28	13	13	10	14	
2	Male	-	79	5	32	-	13	220
	Female	-	54	13	20	-	4	
3	Male	3	9	-	5	-	-	42
	Female	6	14	-	5	-	-	
4	Male	7	5	-	9	-	2	45
	Female	19	1	-	1	-	1	
Total		85	258	36	119	40	66	604

2.3. Procedure

During the final week of August 2025, students beginning their first year in sport-related undergraduate or foundation programmes—both newly enrolled and those who had completed pre-enrolment—received invitations to participate across all four institutions. Following institutional ethical approval processes, relevant student information including full name, gender identity, enrolled programme, and personal email contact details was extracted from post-clearing institutional databases. A secure, password-encrypted file containing this information was transferred to evasys, an external survey administration company responsible for distribution and response collection. Students

received individualized, institutionally branded email messages containing unique survey access links. Upon providing consent, participants received notification regarding their right to withdraw from participation at any point without providing justification, with confidentiality protected through unique identifier assignment. The survey distribution occurred across four waves: an initial distribution during week 1 targeting all enrolled students, followed by three additional distributions during weeks 2, 3, and 4 for newly enrolled first-year students. Survey collection concluded on the Friday preceding each institution's welcome week activities.

2.4. Design and Analysis

A quantitative pre-arrival survey (PAS) was used to explore pre-arrival confidence and perceptions among undergraduate students across four institutions. The PAS included items on learning, community, employability, and health and well-being, asking participants to rate both confidence and importance on a 5-point Likert scale (see [Hibbs et al. 2025a](#) for further details). Demographic information, including prior educational background and first-generation status, was also collected. The PAS is available from the first author.

2.5. Analysis Strategy

Descriptive statistics (means and standard deviations) were calculated for overall PAS scores and the eight subscales (learning confidence, LC; learning importance, LI; community confidence, CoC; community importance, CoI; employability confidence, EC; employability importance, EI; health and well-being confidence, HC; health and well-being importance, HI), disaggregated by gender and programme category across the combined institutional dataset (presented in Table 3). Data screening preceded analysis to verify normality assumptions (via Levene's test) and identify potential outliers. Initial univariate analysis of variance (ANOVA) examined whether socio-economic factors (free school meal eligibility; first-generation student status) and prior college qualifications (qualification type at sixth form/college) influenced responses. Subsequently, we conducted ANOVA with gender ($n = 2$), programme category ($n = 6$), and institution ($n = 4$) serving as independent variables and total PAS score as the outcome variable. A multivariate analysis of variance (MANOVA) followed, utilizing gender, programme category, and institution as independent variables with the eight subscales (LC, LI, CoC, CoI, EC, EI, HC, HI) as dependent measures. Sidak-adjusted post hoc tests evaluated significant main effects for programme category and interaction terms. We calculated effect sizes using partial eta squared (η_p^2) interpreting 0.01 to <0.06 as small, 0.06 to <0.14 as medium, and >0.14 as large ([Cohen 1988](#)). Where sample sizes permitted ($n > 15$), we employed independent samples t -tests to examine within-programme differences. Cohen's d provided effect size estimates, with <0.2 indicating small, 0.2 to <0.5 indicating medium, and >0.8 indicating large effects ([Cohen 1988](#)). Statistical significance was established at $p < 0.05$. SPSS version 28 processed all analyses.

Table 3. Descriptive statistics (M, SD) for eight subscales across programme categories disaggregated by gender (LC = Learning Confidence; LI = Learning Importance; CoC = Community Confidence; CoI = Community Importance; EC = Employment Confidence; EI = Employment Importance; HC = Health and Well-being Confidence; HI = Health and Well-being Importance; M = Male; F = Female).

	Programme of Study											
	Sport and Exercise Science		Physiotherapy and Rehabilitation		Sport, Exercise, and Nutrition		Sports Coaching		Sports Management		Sport Foundation Year	
	M	F	M	F	M	F	M	F	M	F	M	F
LC	3.13 0.38	3.01 0.39	3.21 0.41	3.17 0.48	3.29 0.29	3.02 0.43	3.09 0.40	3.04 0.37	3.07 0.33	3.24 0.57	3.02 0.50	2.85 0.49
LI	3.37 0.38	3.53 0.41	3.72 0.34	3.65 0.38	3.56 0.48	3.59 0.44	3.33 0.37	3.47 0.47	3.37 0.42	3.60 0.41	3.34 0.44	3.34 0.40
CoC	2.77 0.51	2.78 0.54	3.00 0.47	2.93 0.61	2.75 0.68	2.96 0.52	2.76 0.53	2.83 0.36	2.87 0.50	2.10 0.59	2.83 0.47	2.58 0.43
CoI	3.12 0.45	3.17 0.44	3.30 0.42	3.23 0.60	3.15 0.60	3.28 0.47	3.06 0.44	3.20 0.45	3.20 0.61	3.32 0.35	3.12 0.47	3.21 0.44
EC	3.09 0.49	3.02 0.61	3.23 0.47	3.14 0.46	3.26 0.45	3.14 0.46	3.05 0.52	3.12 0.44	3.14 0.51	3.17 0.62	3.02 0.53	2.77 0.52
EI	3.14 0.41	3.20 0.37	3.27 0.39	3.13 0.41	3.13 0.50	3.28 0.50	3.07 0.40	3.09 0.44	3.09 0.47	3.03 0.44	3.17 0.40	3.07 0.45
HC	3.20 0.43	3.14 0.50	3.40 0.41	3.23 0.52	3.02 0.47	3.21 0.54	3.13 0.45	3.20 0.51	3.25 0.48	3.18 0.55	3.23 0.45	2.89 0.65
HI	3.19 0.66	3.35 0.53	3.47 0.45	3.36 0.61	2.90 0.61	3.47 0.59	3.21 0.61	3.50 0.59	3.18 0.65	3.57 0.45	3.36 0.66	3.25 0.46

We performed separate confirmatory factor analyses (CFA) for the Confidence and Importance domains to evaluate the PAS's psychometric characteristics. AMOS 26 (Arbuckle 2003) conducted CFA using maximum likelihood estimation, evaluating model fit through multiple indices: (a) Chi-square statistic ($p > 0.05$ suggests acceptable fit, though sample size influences this metric; larger samples may yield significant results despite minimal data-model discrepancies) (Anderson and Gerbing 1988); (b) goodness of fit index (GFI) and comparative fit index (CFI) (values > 0.95 indicate good fit, > 0.90 indicate acceptable fit) (Hu and Bentler 1999); (c) root mean square error of approximation (RMSEA) (values ≤ 0.06 represent good fit, values > 0.06 and ≤ 0.08 represent acceptable fit) (Browne and Cudeck 1993); (d) Pclose (should be non-significant) (Hu and Bentler 1999; Hooper et al. 2008). Model modifications based on modification indices addressed any inadequate initial fit. Error term correlations were limited to within-factor items due to their theoretical similarity (Byrne 2016). Internal consistency was evaluated through Cronbach's alpha at both domain and subscale levels.

3. Results

3.1. CFA

An acceptable fit for Importance was obtained whereas for Confidence, several cross-correlations of error terms within factors were required to obtain an acceptable fit (see Table 4). The Chi square was significant for both domains which was likely due to the relatively large sample size ($n = 603$). The GFI and CFI had an acceptable value for both domains whereas the RMSEA had a good fit and the PClose was significant. For the Confidence model, we cross-correlated the error terms of the following items: Learning 6–7; Community 1–2 and 2–6; Employability 3–4 and 6–7; Health 1–2 and 1–6. Reliability analysis showed that for Confidence alpha was excellent and good for importance. For the individual factors alpha was acceptable to good (LC = 0.78; CoC = 0.81; EC = 0.88; HC = 0.82; LI = 0.79; CoI = 0.75; EI = 0.74; HI = 0.85). Overall, this indicates that the PAS has acceptable reliability and factorial structure.

Table 4. Confirmatory factor analysis model fit statistics for Confidence and Importance domains of the PAS.

Fit Indicator	Confidence	Importance
Chi Square	791; $p < 0.001$	387; $p < 0.001$
GFI	0.907	0.932
CFI	0.918	0.939
RMSEA	0.054	0.052
Pclose	0.06	0.26
Cronbach α	0.91	0.86

GFI = goodness of fit index, CFI = comparative fit index; RMSEA = root mean square error of approximation; Pclose = P of the close fit.

3.2. Socio-Economic Status and Previous Qualifications

The ANOVA did not show a significant effect of free school meals ($F(6,569) = 1.12$; $p = 0.35$; $\eta_p^2 = 0.01$), first-generation student ($F(6,569) = 1.16$; $p = 0.33$; $\eta_p^2 = 0.01$), or previous college qualification ($F(6,569) = 1.05$; $p = 0.39$; $\eta_p^2 = 0.01$) on total PAS score.

3.3. Pre-Arrival Total Score

Analysis of variance examining the overall PAS scores revealed a statistically significant effect for programme category ($F(5,604) = 3.98$; $p = 0.01$; $\eta_p^2 = 0.03$). However, neither gender ($F(1,604) = 0.01$; $p = 0.96$; $\eta_p^2 = 0.01$) nor institutional affiliation ($F(3,604) = 0.76$,

$p = 0.52$; $\eta_p^2 = 0.01$) demonstrated significant main effects. Additionally, the interaction between programme and gender ($F(1,604) = 1.60$; $p = 0.16$; $\eta_p^2 = 0.01$) and the interaction between programme and institution ($F(1,604) = 1.86$; $p = 0.06$; $\eta_p^2 = 0.03$) failed to reach statistical significance. Post hoc testing using Sidak adjustment for programme category indicated that students enrolled in physiotherapy and rehabilitation achieved significantly elevated scores (3.23 ± 0.35) relative to those in sports coaching (3.09 ± 0.30) and foundation year programmes (3.06 ± 0.31). It should be noted that elevated scores reflect greater confidence levels and heightened perceived importance. Examining differences within specific programme categories, a significant variation emerged for sport, exercise, and nutrition students across institutions 1 and 2 ($T(34) = 1.94$; $p = 0.05$, Cohen's $d = 0.65$). Additional within-programme comparisons across institutions yielded no further significant differences.

3.4. Pre-Arrival Factors

Multivariate analysis of variance demonstrated statistically significant main effects for programme category (Wilks' $\lambda = 0.87$; $p < 0.001$; $\eta_p^2 = 0.03$), gender (Wilks' $\lambda = 0.96$; $p = 0.01$; $\eta_p^2 = 0.04$), and institutional affiliation (Wilks' $\lambda = 0.91$; $p < 0.01$; $\eta_p^2 = 0.031$). Conversely, the interaction between programme and gender (Wilks' $\lambda = 0.92$; $p = 0.42$; $\eta_p^2 = 0.02$) and the interaction between programme and institution (Wilks' $\lambda = 0.86$; $p = 0.31$; $\eta_p^2 = 0.02$) did not achieve statistical significance. Follow-up univariate analyses are detailed in Table 5.

Table 5. Follow-up ANOVA results examining main effects of gender and programme category, and interaction effects between programme and gender, and programme and institution (LC = Learning Confidence; LI = Learning Importance; CoC = Community Confidence; CoI = Community Importance; EC = Employment Confidence; EI = Employment Importance; HC = Health and Well-being Confidence; HI = Health and Well-being Importance). * Denotes statistical significance.

	Institution			Gender			Programme of Study			Programme $\times$ Institution Interaction			Programme $\times$ Gender Interaction		
	F(3,604)	p	η_p^2	F(1,604)	p	η_p^2	F(5,604)	p	η_p^2	F(1,604)	p	η_p^2	F(1,604)	p	η_p^2
LC	1.59	0.19	0.01	2.12	0.15	0.01	1.80	0.06	0.02	1.96	0.05	0.02	1.04	0.40	0.01
LI	4.35	0.01 *	0.02	3.38	0.04 *	0.01	6.38	<0.001 *	0.06	0.89	0.53	0.1	1.48	0.20	0.01
CoC	1.89	0.13	0.01	0.28	0.60	0.01	3.15	0.03 *	0.02	1.60	0.12	0.02	1.30	0.26	0.01
CoI	0.70	0.55	0.01	2.11	0.15	0.01	1.67	0.19	0.01	1.10	0.38	0.02	0.47	0.80	0.01
EC	0.06	0.98	0.01	1.48	0.22	0.01	2.41	0.10	0.02	2.00	0.05	0.03	0.76	0.58	0.01
EI	2.53	0.06	0.01	0.01	0.80	0.01	1.26	0.19	0.01	1.39	0.20	0.02	1.10	0.38	0.01
HC	0.68	0.56	0.01	1.29	0.26	0.01	1.15	0.42	0.01	1.52	0.15	0.02	1.84	0.10	0.02
HI	0.96	0.41	0.01	8.48	0.01 *	0.02	1.46	0.56	0.01	1.02	0.42	0.01	2.43	0.03	0.02

Regarding gender effects, female participants reported significantly elevated scores on LI and HI relative to male participants (refer to Table 3). Post hoc testing for programme category revealed that concerning LC, physiotherapy students obtained significantly elevated scores compared to sport foundation students. For LI, physiotherapy students demonstrated significantly higher scores than students in sport and exercise science, sport coaching, foundation year, and sports management programmes. Examining CoC, physiotherapy students achieved significantly greater scores than sport and exercise science students (all $p < 0.05$). Institutional comparisons indicated that LI scores at Institution 4 were significantly elevated compared to the remaining three institutions ($p < 0.05$; Table 6). Within-programme analysis revealed differences for sport, exercise, and nutrition students regarding EC ($T(34) = 2.32$; $p = 0.03$, Cohen's $d = 0.77$) and EI ($T(34) = 2.68$; $p = 0.03$, Cohen's $d = 0.89$), with Institution 2 demonstrating higher scores (EC 3.34 ± 0.47 ; EI 3.44 ± 0.48) in comparison to Institution 1 (EC 3.00 ± 0.38 ; EI 3.03 ± 0.43). It should be noted that elevated scores correspond to greater confidence and heightened perceived importance.

Table 6. Descriptive statistics (M, SD) for eight subscales by institutional affiliation (LC = Learning Confidence; LI = Learning Importance; CoC = Community Confidence; CoI = Community Importance; EC = Employment Confidence; EI = Employment Importance; HC = Health and Well-being Confidence; HI = Health and Well-being Importance).

Institution of Study	LC	LI	CoC	CoI	EC	EI	HC	HI	Overall
1	3.08 0.42	3.44 0.40	2.86 0.51	3.17 0.47	3.08 0.49	3.10 0.40	3.18 0.48	3.30 0.62	3.13 0.31
2	3.10 0.40	3.43 0.43	2.75 0.52	3.13 0.47	3.08 0.54	3.18 0.43	3.17 0.51	3.25 0.61	3.12 0.31
3	3.03 0.41	3.42 0.44	2.83 0.54	3.23 0.50	3.06 0.57	3.18 0.42	3.21 0.48	3.42 0.61	3.14 0.35
4	3.21 0.46	3.66 0.37	2.79 0.57	3.16 0.55	3.10 0.47	3.22 0.41	3.28 0.40	3.33 0.55	3.20 0.33
Overall	3.10 0.41	3.45 0.41	2.82 0.52	3.16 0.48	3.08 0.52	3.14 0.41	3.19 0.48	3.30 0.61	

4. Discussion

This study pursued two aims: firstly, it investigated pre-arrival confidence and the perceived significance of these factors among first-year sport students attending three post-92 universities and one public research university with a focus on differences related to gender and academic programmes. The second aim was to test the pre-arrival survey's measurement invariance and psychometric properties across different sport degree programs and institutions. The results offer valuable insights into the transitional experiences of sport students entering HE and substantiate the reliability and factorial structure of the PAS for evaluating pre-arrival confidence and expectations within diverse educational settings.

4.1. Programme of Study, Gender, and Institution Differences

Students rated the importance of learning higher than the other three themes, with employability receiving the lowest rating. This indicates that students prioritise knowledge acquisition and academic skill development over immediate career outcomes. Additionally, students expressed greater confidence in their health and well-being competencies compared to their community engagement abilities, with community confidence being the lowest among all eight factors assessed. These results highlight the need for HE institutions to offer activities and resources aimed at strengthening students' community engagement skills prior to their arrival. Insufficient community confidence may lead to reduced participation in group activities and challenges in developing a sense of belonging, thereby increasing the risk of student withdrawal.

The observation that physiotherapy and rehabilitation students consistently exhibit higher levels of pre-arrival confidence and perceived importance compared to students enrolled in sport programmes is consistent with prior research (Hibbs et al. 2025a). These inter-programme differences may be attributable, in part, to the more structured and professionally regulated framework of physiotherapy education, which tends to feature clearly articulated entry criteria, adherence to professional standards, and well-defined career pathways (Shen et al. 2024; Wong and Kaur 2017). Consequently, physiotherapy students often demonstrate greater clarity regarding academic expectations and frequently engage in subject-specific preparation before commencing their studies, thereby enhancing their readiness to meet university requirements. Furthermore, engagement with professional networks or communities prior to arrival may contribute to their elevated sense of belonging, as supported by increased community confidence scores. In contrast, students pursuing broader sport degrees tend to encounter increased ambiguity surrounding programme expectations, potential career trajectories, and academic demands (Timmis et al. 2024; Hayman et al. 2025). Moreover, these individuals may feel compelled to obtain supplementary qualifications—such as those offered by sporting national governing bodies—and seek

additional work experience beyond the scope of their academic curriculum to enhance post-graduation employability (Hall et al. 2019). The current findings indicate that such uncertainty manifests prior to university entry, potentially influencing subsequent levels of student engagement and retention.

The finding that female students reported significantly higher scores regarding the importance of learning and health and well-being compared to male students corroborates and builds upon the findings of Hibbs et al. (2025a), indicating that gender differences in pre-arrival confidence and perceived importance are evident across several UK HE institutions. The elevated value placed on learning by female students may partially account for their higher rates of degree completion and academic achievement within sport programmes (Tomlinson et al. 2024; Hibbs et al. 2025b), while also reflecting a greater investment and stronger motivation towards university engagement. Such increased engagement may facilitate a smoother and more successful transition into HE. Although female students rated learning and health and well-being as more important, there were no significant gender differences observed in confidence domains. This partially aligns with Hibbs et al. (2025a), who reported similar confidence levels in three out of four themes (learning, employability, and health). These findings suggest that while male and female students demonstrate comparable confidence in their capacity to succeed upon entering HE, they differ in the importance attributed to various aspects of the university experience.

The fact that male students place less importance on health and well-being is concerning, especially since research shows clear connections between healthy habits, academic involvement, and student retention (Thomas et al. 2021). Studies have shown that these differences are influenced by gendered attitudes toward seeking help and certain social expectations. For instance, Brown et al. (2021) discovered that students at six UK universities who endorsed traditional masculine ideals—such as self-reliance, mastery, and emotional restraint—were much less likely to seek academic support or express their well-being needs. On the other hand, female students tended to be more open to asking for help and building relationships with staff, which has been associated with better workload management and easier transitions into university life. These patterns indicate that societal norms about how men and women should handle help-seeking and independence may lead male sport students to have lower awareness of health issues and less emphasis on well-being, perpetuating gender gaps in academic achievement and retention rates (Bradburn et al. 2025). The lack of any notable differences between genders across various sports programmes suggests that these trends are widespread rather than limited to specific types of courses. This indicates a need for sector-wide rather than programme-specific approaches to address the gender differences in pre-arrival confidence and perceived importance.

The finding that Institution 4 reported substantially higher learning importance scores than the other three institutions requires thoughtful analysis. This result should prompt further investigation into the institutional factors influencing pre-arrival perceptions. Notably, Institution 4's sample included a significantly greater proportion of physiotherapy and rehabilitation students ($n = 26$, 58%), compared to the other institutions (which ranged from 0 to 21%). Since physiotherapy students generally exhibited higher learning importance scores than those in other sport programmes, the observed institutional differences are likely, at least partially, attributable to this sample composition. These findings underscore the necessity of accounting for programme-specific characteristics when evaluating variations at the institutional level.

The differences found between sport, exercise, and nutrition students from Institutions 1 and 2—particularly in employability confidence and perceived importance—suggest that factors such as institutional communication, partnerships, and location may influence employability-related attitudes before arrival. The observed variation might also stem

from programme marketing materials focused on career outcomes, distinct approaches to communicating placement opportunities, or how employability is integrated into the curriculum. To understand these disparities, further qualitative research could be beneficial. However, it is important to interpret these results cautiously because of the small sample size ($n = 36$) and the absence of similar differences among other programme groups, implying that this effect may be unique to this specific programme rather than widespread across sport-related courses. In contrast, no institutional differences appeared in larger programme cohorts (e.g., SES $n = 258$; sports coaching $n = 119$), showing that students across these programmes generally have similar levels of confidence and perceived importance regardless of institution. This consistency strengthens the generalisability of the survey results and supports implementing common frameworks and interventions to aid student transitions into these areas.

A key observation worthy of discussion is the lack of statistically significant effects of socio-economic status indicators on pre-arrival survey responses. Variables such as receipt of free school meals, first-generation student status, or previous qualification did not significantly affect students' confidence levels or perceived importance scores before arrival. This finding is noteworthy and positive, indicating that individuals from varied socio-economic backgrounds commence university with similar levels of self-assurance concerning academic, social, employability, and health-related challenges. Although existing research has extensively documented the impact of socio-economic disadvantage on educational outcomes and students' sense of belonging in HE (Strayhorn 2023; Williams and Roberts 2023), the results of this study suggest that such disparities are more likely to manifest during the university experience, potentially due to differences in support, resources, or community integration after enrolment. From an operational perspective, these findings imply that thoughtfully designed pre-arrival transition support can be equally beneficial for all students. Nonetheless, it underscores the necessity for institutional support systems, teaching strategies, and campus environments to avoid inadvertently disadvantaging students from lower socio-economic backgrounds post-arrival, given that initial confidence or perceptions do not account for subsequent disparities.

4.2. Psychometric Metrics of Pre-Arrival Survey

A key contribution of this study is the demonstration that the PAS exhibits acceptable to excellent psychometric properties across multiple UK institutions and diverse sport programme types. The CFA revealed that both the Confidence and Importance domains achieved acceptable model fit indices. The reliability analysis further supports the robustness of the PAS with values meeting or exceeding conventional thresholds for acceptable internal consistency (Nunnally and Bernstein 1994). The current findings extend the initial work of Hibbs et al. (2025a), who reported acceptable reliability and factorial structure within a single institution. The present findings demonstrate that these properties generalise across institutional boundaries, strengthening confidence in the PAS as a tool for sector-wide assessment of pre-arrival confidence and importance perceptions. This has key implications for national benchmarking efforts and inter-institutional comparisons of student preparedness. The ability to use a standardised instrument across multiple universities enables more systematic evaluation of pre-entry interventions and facilitates identification of institution-specific or programme-specific areas requiring targeted support.

4.3. Practical Implications

The results of this study present significant implications for higher education providers, especially those delivering sport and physiotherapy-related programs. The successful validation of the PAS across multiple institutions offers the sector a dependable, standardized

instrument for evaluating incoming students' confidence and perceptions regarding learning, community engagement, employability, and health and well-being. Institutions may integrate the PAS into their pre-arrival onboarding processes to proactively identify students who could benefit from additional transitional support. Given the survey's robust reliability and factorial structure, benchmarking data can be utilized for sector-wide comparisons, enabling collaborative insights into effective pre-arrival interventions.

Secondly, differences between programmes—especially the lower confidence observed among foundation year and sports coaching students—indicate that tailored interventions are necessary. For these groups, strategies could involve providing more comprehensive academic resources before arrival, sharing clearer information about academic expectations, offering employability-focused courses, and organising community-building activities. Meanwhile, for those in less vocationally oriented programmes such as sport and exercise science, support might include efforts to clarify career options and encourage participation in professional networks. Additionally, since community confidence was rated as the lowest competency, this issue should be prioritised throughout the sector so that all new higher education students receive the support they need for a smoother transition.

Third, differences in how male students value certain aspects suggest they might benefit from communication that highlights why actively participating in both academics and healthy behaviours can improve their success at university and beyond. Before arrival, messaging could be crafted to challenge gender stereotypes and help male students see the benefits of engaging fully with campus life. For example, sharing stories of male students who excel academically, graduates discussing how employability skills helped their careers, or students who improved their performance through health and well-being programs could all illustrate these points.

Given the well-known challenges of retention and progression in UK sport programmes (HESA 2024; Timmis et al. 2022), these findings highlight the need to identify and support students early with respect to their confidence and perceptions. Institutions have the ability to influence students' confidence and sense of importance even before they arrive on campus. By gathering, evaluating, and addressing mismatches in student confidence and perceived importance prior to arrival, universities could lower the likelihood of early disengagement and withdrawal. This approach reflects a move from reactive support—helping students only after issues arise—to proactive preparation, ensuring students begin their studies as prepared and confident as possible.

4.4. Limitations

Although a 58% response rate is strong, non-response bias may affect the results, as survey completers could differ from non-responders in ways that influence confidence scores. The study's focus on three post-92 institutions and one traditional university offers consistent context but limits broader applicability. Differences in student demographics and institutional missions may also impact incoming students' confidence and perceived importance responses compared to those at more selective universities.

The study measures pre-arrival perceptions at one timepoint before university entry. Additional data points are needed to assess changes in these perceptions after arrival. Linking results to student retention and academic performance is essential for developing targeted interventions. While Hibbs et al. (2025b) found that pre-arrival factors predicted academic outcomes at one institution, broader longitudinal studies across multiple institutions are needed to validate the PAS's predictive value.

The grouping of twenty-five individual programmes of study into six programme categories, while necessary for statistical analysis, may obscure important within-category

variation. For example, physiotherapy and sports rehabilitation may attract students with different profiles and pre-arrival perceptions.

4.5. Future Directions

While this study has successfully demonstrated differences in pre-arrival confidence and perceived importance across programmes of study, institutions, and gender, further research is needed to establish acceptable benchmark levels that facilitate a smooth transition into HE. Although each incoming student requires different competencies to succeed—shaped by their previous experiences, current skillset, and motivations—establishing a ‘typical profile’ would enable institutions to develop more targeted and effective interventions.

It is essential to conduct intervention studies assessing whether pre-arrival programmes aimed at enhancing confidence and setting realistic expectations result in measurable improvements in beliefs, retention, and academic performance. Employing experimental designs to compare outcomes between students who participate in pre-arrival support and those who do not will provide causal insights into the efficacy of various intervention strategies. These studies could incorporate pre-arrival modules outlining key expectations and available resources, supplemented with programme-specific case studies. Furthermore, evaluating the comparative effectiveness of programme-specific versus generic institutional support interventions could yield valuable guidance for best practices in student transition support.

The gathering of qualitative research exploring why the observed differences exist would provide valuable insights. Focus groups or interviews with students from different demographic backgrounds, programmes, genders, and institutions would provide knowledge on what shapes pre-arrival confidence and perceived importance alongside what concerns are most common and how universities can most efficiently support pre-arrival preparation. Understanding why male students place lower importance on learning and health and well-being, why students on sport-related programmes have lower confidence than physiotherapy and rehabilitation students, investigating whether pre-arrival confidence matters more for students living away from home would help design more effective targeted interventions.

Extending the study more beyond post-92 universities would establish whether the psychometric properties and the findings can be generalised across the HE sector in the UK. This would enable sector-wide benchmarking and identification of institutional factors associated with better-prepared student cohorts. Extending the reach of the PAS internationally would establish whether the observed trends in pre-arrival confidence and perceived importance in the UK are replicated globally in this student sample. Equally, examining how the PAS score relates to or predicts other validated measures of transition success (such as sense of belonging and academic self-efficacy) would strengthen the validity of the survey.

5. Conclusions

This multi-institutional study makes important contributions to understanding the pre-arrival confidence and perceptions of sport students entering UK HE. The psychometric validation of the pre-arrival survey demonstrates its reliability and factorial validity, providing the sector with a robust, standardised tool for assessing incoming students’ preparedness and early identification of students who may benefit from additional transitional support. The findings reveal significant variation in pre-arrival score by programme and at factor level by programme, gender and institution with physiotherapy students demonstrating consistently higher scores compared to sport-related programmes and female students placing greater importance on learning and health. Practical implications for the sector are

outlined with possible intervention strategies proposed. By implementing the pre-arrival survey systematically and responding proactively to identifying needs, universities can better support student transitions and address the persistent retention and progression challenges facing UK sport programmes.

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Data Availability Statement: The dataset will be made available upon acceptance for publication.

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Notes

- ¹ HE in the UK system refers to post-secondary education provided by universities, colleges or HE institutions.
- ² Post-92 refers to higher education institutions in the United Kingdom who were granted university status through the Further and Higher Education Act 1992. This can include both former polytechnic colleges and institutions that have been created since 1992.

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